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Survival analysis of patients with COVID-19 using deep neural network and random forest techniques

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A B S T R A C T

Introduction: The prediction of the survival chance of coronavirus disease 2019 (COVID-19) patients is as important as the early detection of the coronavirus. Since patient mortality, factors may differ by location, this study concentrated on identifying the influential factors and predicting survival for COVID-19 patients using machine learning methods in Fars province, Iran.

Material and Methods: The research dataset was extracted in the period January 21, 2020, to September 25, 2020, and contains 25858 hospitalized patients' records with 51 features. These records were classified into two categories: death (label 1) and survival (label 0). The methodology of this research is CRISP standard. A comparison was made between the efficiency of two deep neural network and random forest algorithms in predicting survival. Modeling steps were done with Python language in the Google Colab environment.

Results: Experimental results demonstrated that the deep neural network algorithm had better performance than random forest with accuracy, precision, recall, F-score, and receiver operating characteristic of 97.2%, 100%, 93.54%, 96.66%, and 97.9%, respectively. Based on the results of the random forest model, history of hypertension, chronic neurological disorders, chronic lung diseases, asthma, chronic kidney disease and, heart disease were the most important risk factors related to death.

Conclusion: Deployment of our proposed model allows medical professionals to exercise greater caution during the treatment of patients who are most likely to die due to their medical conditions.

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INTRODUCTION

Coronavirus disease in the 2019 outbreak and the world health organization (WHO) in 2020 introduced it as a pandemic. Outcomes caused by COVID-19 can range from symptoms to wider body involvement and, in more acute stages, death. Therefore, categorizing patients based on symptoms to allocate resources is an important step in disease management [1]. In general, various methods such as nucleic acid-based methods using polymerase chain reaction (PCR), computed tomography (CT), and scan chest X-ray (CXR) can be used for COVID-19 diagnosis. Despite their benefits, can have limitations, such as the adverse effects of CT scans on adjacent bones.

Therefore, the use of assistive methods based on artificial intelligence (AI) such as machine learning (ML) and deep learning (DL) can have a significant impact on improving the accuracy and speed of diagnosis [2-4]. ML has become one of the emerging methods in the medical sciences and has had a remarkable result in the COVID-19 diagnosis [5-8]. In addition to diagnosis, these methods are also used to

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determine the survival rate of patients. Survival analysis is a model for determining the amount of time left until a specific event [8] and can be used to predict the length of stay in hospital or the length of time to death [9, 10]. Such analysis helps the clinicians identify critically ill patients more quickly and make better decisions to continue the care process [9].

There is a wide class of survival prediction models, from those built using traditional regression methods to more recently developed ML algorithms [11]. Time-to-event is considered a continuous outcome in traditional approaches. In addition, in these approaches, parametric or semi-parametric assumptions are made about the distribution of survival times. Parametric survival models often assume a specific distribution for survival time based on prior knowledge from the clinical or scientific context. These models can be more accurate when survival times follow an assumed parametric distribution but can direction to biased estimates if misspecified.

The semi-parametric Cox proportional hazards model leaves the distribution of the survival times to be estimated using non-parametric methods but incorporates covariate effects so that individuals with different sets of predictors can have different predicted survival curves. The models based on standard Cox rely on a proportional hazards' hypothesis for the survival times that, when violated, can lead to wrong predictions [12].

ML algorithms are designed to get optimal predictive performance, do not require auxiliary relationship specifications, and are capable of capturing complex and nonlinear relationships in data. As a distinction between parametric and semi-parametric survival models and ML models, they do not make distributional or proportional hazard assumptions about survival time. The disadvantages of survival prediction models based on ML include low interpretability, overfitting, and computational burden [12, 13].

In previous studies, various tools based on AI predicted deaths caused by COVID-19. Clinical decision support systems (CDSSs) are among the ones that, relying on ML, were able to improve treatment outcomes for patients with COVID-19. The results of the study by Shanbezadeh et al., which was based on the design of a CDSS to predict the risk of mortality in patients with COVID-19, showed that BP-ANN, with the best performance among other models, is able to better predict cases of COVID-19 deaths and increase safety [14].

In addition to DSSs, registry systems also play an important role in standardizing the data sources of COVID-19 patients and thus creating a common platform for prognosis, treatment plan complications and survival estimation [15]. ML models can used as a basis for COVID-19 prediction mortality. Moulae et al. predicted the death rate of patients with COVID-19 based on ML algorithms, and in the end, the random forest (RF) algorithm with an accuracy of 95.03% had the best performance among another model [1]. These ML models are able to integrate with various tools such as health information systems, mobile health applications, expert systems and other tools in the health system to improve the accuracy of health decisions.

Various studies have analyzed the survival of patients with COVID-19 using DL methods. Atlam et al. developed Cox_COVID_19 based on Cox regression and Deep_Cox_COVID_19 with concordance of 0.983 for Deep_Cox_COVID_19 system and 0.923 for Cox_COVID_19 system and concluded both models can predict survival probability and provide the most important features for distinguishing death and severer patients [16]. Sankaranarayanan et al. according to electronic health record and laboratory information, developed GRU-D recurrent neural network which achieved the highest peak cross-validation performance with 0.938 (SE=0.004) between other models and can predict the mortality probability in the first 72 hours after positive PCR test [17].

In addition, Khozeimeh et al. propose novel model CNN- auto encoders (AE) to predict survival chances according to clinical information and CT images. Dataset was balanced by using a data augmentation procedure based on AEs. They concluded that CNN trained on clinical data could perform as well as those, which are trained by CT images. The evaluation factors were 95% confidence interval (CI) for the accuracy, recall and specificity of the CNN-AE 96.05±1.48%, 98.00±1.33% and 93.13±2.52%, respectively [18].

Näppi et al. developed a survival prediction model based on CT images named U-survival that integrates chest scan DL and elastic-net Cox survival model and had performance of maximum concordance index: 91.6% [95% CI 91.5, 91.7] [19]. Shu et al. propose a DL model based on time to event analysis, which incorporates images, and non-images study into time to event analysis. Their results show that image sequence can significantly promote survival prediction. Their method had concordance error of around 25% based on clinical data and 20% for X-rays based on analysis for predicting hospital admission [20].

Moulae et al. in their study conclude that RF algorithms with accuracy, sensitivity, precision, specificity, and receiver operating characteristic (ROC) of 95.03%, 90.70%, 94.23%, 95.10%, and 99.02%, respectively had the best performance between other algorithms and dyspnea, intensive care unit (ICU) admission, and oxygen therapy were

best predictors for death predictions [1]. Wang et al. applied the RF algorithm predict outcomes in COVID-19 patients and by performed recursive feature elimination (RFE), 11 clinical factors were chosen for analysis which lactate dehydrogenase (LDH) and myoglobin (Myo) based on Gini index were selected as optimal predictors which the area under the curve (AUC) for test data was 100% [21].

Sharma et al. based on an improved RF (IRF) algorithm to propose a model to predict COVID-19 patients' health and conclude that IRF perfume better in terms of different index such as precision (0.765), ROC area (0.757) in comparison with RF and support vector machines (SVM) algorithms. Number of hospitals, ventilators, beds and treatments facility were most important factors, which predict patient's health outcomes [22]. Also, Gupta et al. by using observation date, time and state territory as the most important features and algorithms such as RF, SVM, neural networks, decision tree (DT) and multinomial logistic regression estimate deaths, confirmed and cured cases of COVID-19 in different territory of India. They conclude that RF with better accuracy outperformed the other algorithms [23].

In the previous studies, the mentioned demographic, clinical, laboratory, and therapeutic factors affect the survival rate and can decrease or increase them. However, due to differences in the behavior of the coronavirus in different countries, the effect of these risk factors on recovery or mortality may be different. Our goal is to provide a survival prediction model with acceptable accuracy and determine the effective factors in the death of COVID-19 patients in Fars province, Iran. The results of this research can help the medical community to timely detection of the mortality risk of COVID-19 patients, which is necessary to improve hospital management systems and resource allocation standards. This study presents a number of contributions as follows: The survival prediction model of COVID-19 patients based on clinical features. An investigation of the effects of clinical features on the COVID-19 mortality.

MATERIAL AND METHODS

The research methodology is based on the cross industry standard process for data mining (CRISP-DM) standard. The CRISP-DM methodology provides an overview of the life cycle of a data-mining project. It is a process model that serves as the base for a data science process and contains the phases of a project, their respective tasks, and the relationships between them (Fig 1). All methods of the present study were performed in accordance with the relevant guidelines and regulations. Protected personal health information was removed for this research.

All the research method phases were implemented using Keras-Pandas [24, 25]. It allows users to rapidly build and iterate on DL models. This is a Python DL application programming interface (API). The CRISP-DM has six sequential phases.

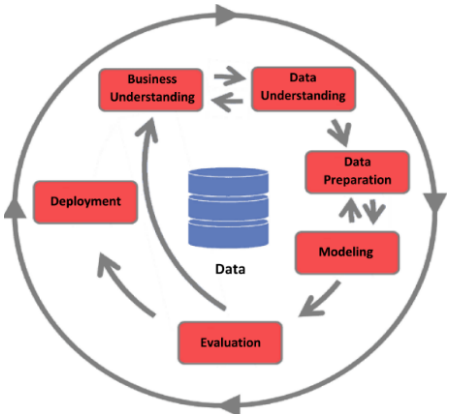


Fig 1: Phases of the CRISP-DM reference model [26]

Each phase of the CRISP-DM method is described below:

Business understanding

This initial phase focuses on the project objectives and requirements. Due to the increase in the number of COVID-19 patients and the lack of equipment to accommodate all patients, a difficult choice has been made. Necessary medical care is applied only to patients who are most likely to survive. The effect of related risk factors on recovery or mortality may vary from country to country [25]. In Iran, from 3 January 2020 to 5:54 pm to 29 July 2022, there have been 7,368,945 confirmed cases of COVID-19 with 141,837 deaths, reported to WHO [24]. Considering the high mortality rate of this disease in Iran. This research goal is to provide a survival prediction model and determine the effective factors in the death of COVID-19 patients.

Data understanding

This phase begins with initial data collection and continues with activities to become familiar with the data, identify data quality problems, and discover first insights into the data. This retrospective hospital-based study was conducted on suspected people who were referred to the COVID-center hospitals and examined by an infectious disease specialist. Finally, based on the results of the PCR test or CT scan, they were classified into two infected and non-infected classes. Based on the severity of the disease, according to the infectious disease specialist opinion, the need for hospitalization of the patients has been diagnosed.

Initial data includes demographic characteristics (such as age and gender), clinical symptoms, and comorbidities (such as the history of hypertension, diabetes, chronic heart disease, lung disease, and

cancer) were collected from the medical care monitoring center (MCMC) in Fars province. This system has a multi-center database. The research dataset was extracted in the period January 21, 2020 to September 25, 2020 and contains 25858 hospitalized patients' records with 51 features. These records were categorized into two classes, death (label 1), and survival (label 0). At the end of this phase, after exploring the data set, the data quality (how clean/dirty the data is) was verified.

Data preparation

This phase covers all processes to build the final dataset from the initial raw data. First, we removed some features that do not need to be included in the survival prediction model, such as the name of the hospital, city of residence, patents' ID, date and time of registration, date of death. We also removed the records with the null attributes. Finally, the number of features was reduced to 38 and the number of instances was reduced to 20599. In the dataset, the number of instances from the death class (minority) is much smaller than the number of instances of the survival class (majority). We used the SVM algorithm to balance the dataset [27]. In this algorithm, a score is assigned to each data, which is a measure of outliers. A threshold is considered for data deletion and data with a higher score is kept. We removed scores less than 2435; this is a threshold value, which is considered experimental. We removed a class of outlier data within the survival class, and in this way, the dataset was balanced at the same time as the removal of outlier data. After this phase, the total number of instances reached 3053. The numbers of survival and death instances were reduced to 1600 and 1453, respectively. All features were normalized using an automatic scale known as standardization or z-transformation (1).

$$x' = \frac{(x - \text{mean})}{\text{Standard Deviation}(X)} \quad (1)$$

Different feature selection methods are available to determine the weight of each feature in the classification. To determine the effective factors in the death of patients with COVID-19, we used the RF algorithm. RF is a powerful statistical classifier that combines the output of multiple decision trees to reach a single result. The main advantages of RF compared to other classifiers consist of: 1) very high classification accuracy; 2) a powerful method for determining variable significance; 3) ability to model complex interactions between predictor features; 4) flexible to perform several types of statistical data analysis, and 5) an algorithm for handling missing values [28]. RF classification of Scikit-Learn ML in Python [29] was applied to the dataset.

Modeling

In this step, different modeling techniques are selected and applied, and their parameters are calibrated to optimal values. Based on the scope and purpose of the research, different techniques such as clustering, classification, prediction or improvement of results can be used. For example, to classify data into predefined groups, various classification techniques such as DT or SVM can be used. In this research, DNN and RF classification techniques were used. These two algorithms are used as leading techniques in the classification of tabular and structured data, and since the data of the current research was structured, these two techniques were used [30, 31].

Typically, there are several techniques for the same data mining problems, such as placing data in predetermined classes or finding a group of objects that have common characteristics or predicting the occurrence of an outcome in the future. Some techniques have specific requirements on the data form. Therefore, it is often necessary to return to the data preparation stage.

DL techniques can be used for COVID-19 survival analysis [16]. One of the most prominent DL techniques is DNN. It is a powerful model with a wide range of applications [32]. It is an artificial neural network with several layers between input and output. Therefore, DNN consists of an input layer, n-hidden layer and an output layer where each layer consists of nodes. While the input layer receives the input data, the hidden layers perform mathematical calculations on the inputs. The number of nodes in the input and output layer, bias, learning rate, initial weights to adjust, number of hidden layers, number of nodes in each hidden layer, and stopping condition to end cycles during execution are the basic parameters used in DNN [25, 33].

In this study, a deep network has four layers. The first layer has 27 neurons and the rectified linear unit (ReLU) activation function. The second layer has 18 neurons and the ReLU activation function. The third layer has nine neurons and the ReLU activation function. In the last layer, the research used the sigmoid function. Adam optimization algorithm was used, because this optimizer gave the best result for this research problem, and in this network and the training rate was equal to 10^{-4} . This optimizer is an extension to stochastic gradient descent. It should be noted that the loss function in this optimization was binary cross entropy. The training of the weights was done through the stochastic gradient descent (SGD) algorithm and the number of epochs was equal to 30. During the training, we considered 20% of the data validation data to avoid overtraining. The used DNN model is illustrated in Fig 2.

This research also used the RF algorithm for predicting survival and compared its performance with the DNN model. RF implementation details are

illustrated in Table1.

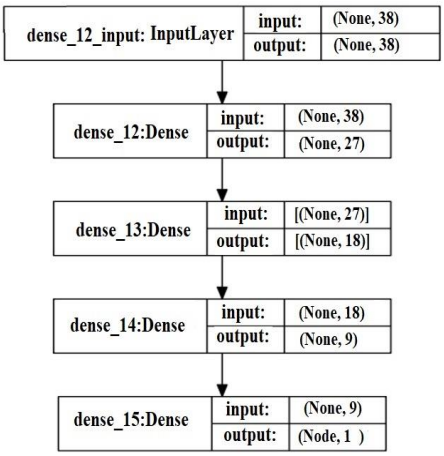


Fig 2: Proposed DNN architecture

Table 1: Parameters of random for RF in Python

Parameter	Description	Value
N_estimators	The number of trees in the forest	100
Min_samples_split	The minimum number of samples required to split an internal node	2
Min_samples_leaf	The minimum number of samples required to be at a leaf node	1
Min_weight_fraction_leaf	The minimum weighted fraction of the sum total of weights (of all the input samples) required to be at a leaf node.	0
Criterion	The function to measure the quality of a split	Gini
Max_depth	The maximum depth of the tree	Nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples
Max_features	The number of features to consider when looking for the best split max_features=sqrt(n_features)	sqrt
Max_leaf_nodes	Grow trees with max_leaf_nodes in best-first fashion. Best nodes are defined as relative reduction in impurity	unlimited number of leaf nodes
Min_impurity_decrease	A node will be split if this split induces a decrease of the impurity greater than or equal to this value.	0
N_jobs	The number of jobs to run in parallel	1

To split the dataset, we used the train_test_split() class from scikit-learn library. Dataset were divided into training: testing ratio 75:25.

Evaluation

The evaluation phase looks more broadly at which model best fits the business. The classifier model performance was evaluated in sensitivity, precision, F-measure, and accuracy [34] (Table 2). These metrics are calculated as shown in (2) to (5).

Table 2: Confusion matrix and definitions of terms used for measuring models' performance

		Real class	
		Death	Survival
Predicted class	Death	True Positive (TP): The number of death case that are properly categorized as death	False Positive (FP): The number survival case that are wrongly categorized as death
	Survival	False Negative (FN): The number death case that are wrongly categorized as survival	True Negative (TN): The number of survival case that are properly categorized as survival

$$\text{Sensitivity} = \text{TPR} = \frac{\text{TP}}{\text{TP+FN}} \quad (2)$$

$$\text{Accuracy} = \text{FPR} = \frac{\text{TP+TN}}{\text{TP+FP+TN+FN}} \quad (3)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \quad (4)$$

$$\text{F} - \text{Measure} = \frac{2*\text{Precision}*\text{Sensitivity}}{\text{Percision+Sensitivity}} \quad (5)$$

In addition to the above evaluation criteria, researchers used ROC and AUC [35, 36] to evaluate the advantages and disadvantages of the survival model. The ROC curve shows the trade-off between TPR and FPR 33. This curve plays an important role in evaluating the diagnostic power of the tests to distinguish the real situation and find the optimal cut-off values. One of the reasons for the popularity of this technique is the non-distortion of its results due to the use of decision criteria or cuts. The area under the curve determines the inherent ability of the test to discriminate between healthy and diseased populations and has therefore become a popular method for assessing the accuracy of medical precision diagnostic systems [36]. To split the dataset, we used the train_test_split() class from scikit-learn library. Dataset were divided into training: testing ratio 75:25.

In sum and based on Fig 3, the research included these stages: the data related to COVID-19 patients in Fars province, which included 25,858 records of hospitalized patients and 51 features, was collected. In the next step, unnecessary features such as center name and records with null values were removed.

Then, using the SVM algorithm, the database was balanced, and at the same time, outlier data were removed. After normalizing the features, the effective factors were extracted using the RF algorithm. In the modeling stage, the performance of DNN and RF algorithms was compared by splitting the data into two groups of 25% and 75%. In the evaluation phase, the performance of two algorithms was compared using sensitivity, precision, F-measure, and accuracy factors. To improve the performance of the algorithms, it returned to the preprocessing stage and the modeling was repeated. By achieving the best performance of the algorithms, the final results were introduced to help decision making.

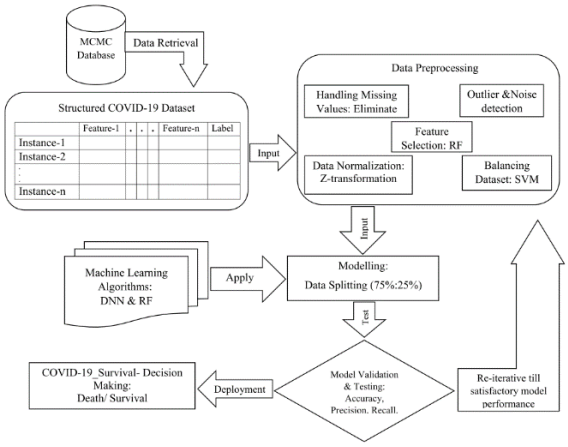


Fig 3: Main block-diagram of proposed approach

RESULTS

This section provides results on the data-driven survival prediction of COVID-19 patients. All features that were used in modeling can be seen in Table 3. For the 2289 training-set instances, 1177 (51.42%) were survival and 1112 (48.58%) were death. For the 764 test-set instances, 423 (56.85%) were survival and 341 (45.83%) were death.

Table 3 demonstrates that, the accuracy of the DNN model on the test-set data is 97.12%. Other evaluation parameters, including precision, recall, and F-score were calculated in the order of 100% and 93.54%, and 96.66%.

Table 3: DNN confusion matrix for test-set

		Real class	
		Death	Survival
Predicted class	Death	TP=319	FP=0
	Survival	FN=22	TN=423

Table 4 shows that, the accuracy of the DNN model on the training-set data is 97.46%. Other evaluation parameters, including precision, recall, and F-score were calculated in the order of 100% and 94.78%, and 97.32%. The area under the ROC curve was calculated 97.9% and 98.5% for test and training-set respectively (Fig 4).

As the RF values in Fig 5 show, there is a strong relationship between the mortality rate of COVID-19 patients and the presence of other important diseases such as heart disease, kidney disease, lung diseases, asthma, neurological disorders and hypertension. Conversely, milder symptoms and/or diseases, such as age, gender, contingence, fever, cough, and muscular pain are less likely to contribute to mortality. For more details, see Table 5.

Table 6 illustrates that, the accuracy of the RF model on the test-set data is 96.33%. Other evaluation parameters, including precision, recall, and F-score were calculated in the order of 92.34% and 100%, and 96.01%.

Table 7 shows that, the accuracy of the RF model on the training-set data is 97.50%. Other evaluation parameters, including precision, recall, and F-score were calculated in the order of 94.75% and 100%, and 97.30%. The area under the ROC curve was calculated at 97.2% for both the test and training set (Fig 6).

The area under the ROC curve was calculated at 97.2% for both the test and training set (Fig 6).

DISCUSSION

In this research, using DNN and RF algorithms to achieve the first objective of this research, we tried to predict the probability of survival of hospitalized patients with COVID-19.

Table 4: DNN confusion matrix for training-set

		Real class	
		Death	Survival
Predicted class	Death	TP=1054	FP=0
	Survival	FN=58	TN=1177

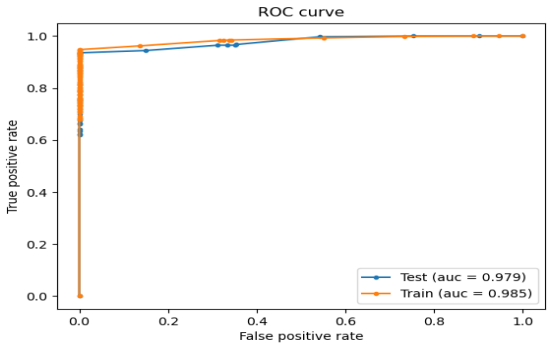


Fig 4: ROC curve of the DNN model in testing dataset

Table 5: The features' importance for death

Features	Score
Age	0
Gender	.00004
Contingence	.00008
Fever	.00012
Cough	.00017
Muscular pain	.00037
Respiratory distress	.00066
Loss of consciousness	.00114
Decreased sense of smell	.00172
Decreased sense of taste	.00182
Convulsions	.00202
Headache	.00260
Vertigo	.00352
Parents	.00373
Pelvic	.00402
Chest pain	.00559
Skin lesion	.00591
Stomach ache	.00628
Nausea	.00656
Vomit	.00805
Diarrhea	.00821
Anorexia	.00929
History of smoking	.01015
Drugs (Opium)	.01089
Po2	.01102
Cancer	.01322
Chronic liver disease	.01772
Diabetes	.02001
Chronic blood diseases	.03985
HIV/AIDS	.04041
Immunodeficiency	.04585
Pregnancy	.04953
Heart disease	.05869
Chronic kidney disease	.06372
Asthma	.07323
Other chronic lung diseases	.09260
Chronic neurological disorders	.09544
History of hypertension	.16979

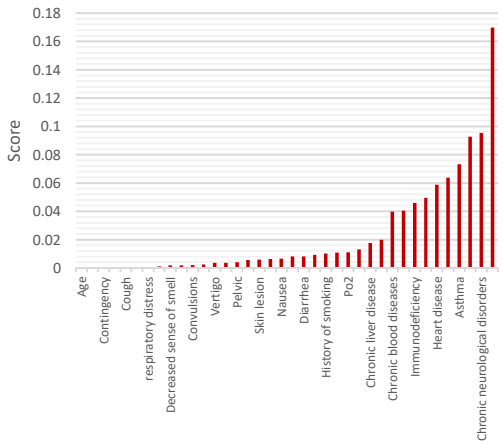


Fig 5: Feature effects on mortality rate based on RF

Table 6: RF confusion matrix for test-set

		Real class	
		Death	Survival
Predicted class	Death	TP=338	FP=28
	Survival	FN=0	TN=398

Table7: RF confusion matrix for training-set

		Real class	
		Death	Survival
Predicted class	Death	TP=1030	FP=57
	Survival	FN=0	TN=1202

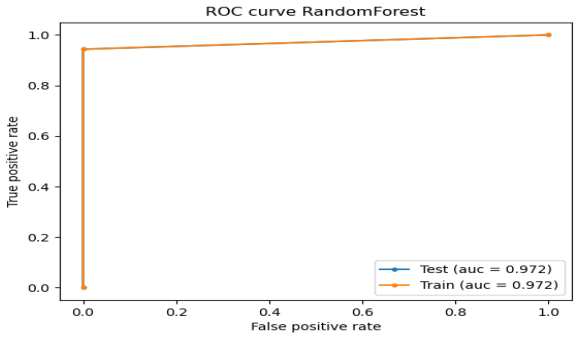


Fig 6: ROC curve of the RF model in testing dataset

The research performed experiments using a clinical dataset. Based on the results, the DNN model has higher accuracy than the RF model.

Using comorbidity and symptoms as input independently, DNN and RF models achieved >97% and >96% predictive accuracy, respectively. As a secondary objective, the role of each risk factor in the survival of patients was evaluated. Input features' importance based on Gini impurity was evaluated. The analysis of the importance of the features is shown in Fig 4. According to Table 5, based on the RF model, the most important factors affecting the death of patients have been reported. According to the effective features, hypertension has the most effect on the probability is known to have deaths caused by COVID-19.

Also, in a study conducted by Chen et al. with the aim of classification of accurate severe versus non-severe COVID-19 clinical type based on RF algorithm, hypertension was identified as the most important factor. In their study, other factors affecting the severity of COVID-19 included diabetes, gender, cardiovascular disease, hypertension, and age [27]. Contrary to this research, in our study, age and gender had a negligible impact on the survival of patients, which is in line with the results of the Meng et al. [8]. Also, in the study of Sankaranarayanan et al. age was identified as the most valuable risk factor

[17]. Furthermore, in Ikemura et al. study, the 2 most influential variables for generating high-performing mortality prediction were reported systolic and diastolic blood pressure, and age[9]. After comparing the results of our study with other studies such as Chen et al. research [27], we found that the history of heart disease plays a role in both the death of patients and the severity of their deterioration.

Research by Khozeimeh et al. proposed a survival chance predictor for COVID-19 patients using clinical features-based CNN. According to their results, age, cancer, heart and kidney diseases were the most important features related to patients' deaths, respectively. In their study, blood pressure and lung diseases were ranked 7th and 8th [18]. Meanwhile, in our study, these two factors are known to be more effective in the death of patients and ranked first and fourth respectively. Also, cancer was calculated as the 13th most effective factor in survival, and compared to this study, it has had low importance in prediction of survival. Others, such as Atlam et al. used Cox_COVID_19 and Deep_ Cox_COVID_19 prediction systems to predict survival. According to their results, age, pneumonia, muscle pain, and throat pain are the most significant risk factors affecting mortality [16].

None of these features were identified as the ten most important features in predicting survival in our model. In the study of Ahmed et al. the characteristics of blood pressure, chronic obstructive pulmonary disease, blood sugar, asthma, chronic kidney disease, obesity, acute respiratory distress syndrome, and pulse oximetry were used as effective factors in developing a predicting survival system [37]. In line with these results, in our study, the important role of blood pressure, kidney disease, and lung problems has been proven in terms of survival prediction.

Also, Cisterna-García et al. in their study identified age, gender, comorbidities such as hypertension, diabetes, obesity and chronic obstructive pulmonary disease (COPD) as the most important predictors for survival of COVID-19 patients, which was in line with our result in the importance of hypertension and lung diseases [29].

To deployment the proposed DNN model in practice, it should be evaluated by medical experts. Once approved, the system can be used to assist experts to analyze the severity status of COVID-19 patients. Therefore, patients with critical conditions can have higher treatment priority than non-critical. Prioritizing the treatment of patients is of great importance when available medical resources are limited. Deployment of our proposed model, experts can study the relationships between patients' medical conditions (e.g., blood pressure and diabetes) and the likelihood of death from COVID-19. This allows medical professionals to exercise greater caution during the treatment of patients who are most likely to die due to their medical conditions.

Limitation and Future Directions

This study is a retrospective study based on data collected from several clinics in one province, so the generalization of the research results may be limited. On the other hand, lack of access to patients' radiographic data was another limitation of the research. Another limitation of our study was the lack of access to the laboratory data of these patients during the data collection period. In this study, only two algorithms were compared, while using more algorithms and comparing them will create more research horizons for the audience. Therefore, for future studies, it is highly recommended to compare other existing algorithms, using various types of clinical, laboratory and radiographic data, as well as a prospective approach in conducting ML studies.

CONCLUSION

This research investigated the possibility of training a DNN on clinical data to predict the survival of COVID-19 patients. Also, compared its performance with the RF algorithm. The performance of the DNN model was evaluated as higher than that RF. The proposed survival prediction models can then be used by healthcare teams to make significant decisions regarding patient care. For example, deciding to increase monitoring frequency or implement particular therapies. The effect of related risk factors on recovery or mortality may vary from country to country. Therefore, this research evaluated the role of clinical data in the survival of hospitalized patients with COVID-19 in Fars province in Iran.

This research can help potential researchers and clinicians in their research and work on the COVID-19. COVID-19 has caused many deaths worldwide. We suggest that symptoms and comorbidities can be used as an initial screening tool for triaging. It is necessary to conduct a prospective study in order to investigate the effect of the proposed model on reducing the mortality rate caused by COVID-19 and the optimal allocation of hospital resources. As a future work, software development based on proposed DNN model will be investigated. Future work research should be conducted to obtain evidence based on which the use of death warning systems can lead to improved patient outcomes.

AUTHOR'S CONTRIBUTION

AY, AF, OA, and LE have read and approved the manuscript. AY and LE contributed equally to the study design. AY and LE prepared the manuscript and revised it critically. AY, AF, OA, and LE were the initiator of the literature search and review, reading, categorizing and analyzing. AY collected research

data and developed the proposed model. LE additionally performed supervisory tasks.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest regarding the publication of this study.

FINANCIAL DISCLOSURE

No financial interests related to the material of this manuscript have been declared.

ETHICS APPROVAL

Shiraz University of Medical Sciences (SUMS) Ethics Board (IR.SUMS.REC.1399.691) approved the study protocol.

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